

Speed-of-Processing and Driving Simulator Training Result in Improved Driving Performance

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Useful field of view, a measure of processing speed and spatial attention, can be improved with training. We evaluated the effects of this improvement on older adults' driving performance. Elderly adults participated in a speed-of-processing training program ($N = 48$), a traditional driver training program performed in a driving simulator ($N = 22$), or a low-risk reference group ($N = 25$). Before training, immediately after training or an equivalent time delay, and after an 18-month delay each participant was evaluated in a driving simulator and completed a 14-mile (22.5-km) open-road driving evaluation. Speed-of-processing training, but not simulator training, improved a specific measure of useful field of view (UFOV®), transferred to some simulator measures, and resulted in fewer dangerous maneuvers during the driving evaluation. The simulator-trained group improved on two driving performance measures: turning into the correct lane and proper signal use. Similar effects were not observed in the speed-of-processing training or low-risk reference groups. The persistence of these effects over an 18-month test interval was also evaluated. Actual or potential applications of this research include driver assessment and/or training programs and cognitive intervention programs for older adults.

INTRODUCTION

Some older drivers are prone to crashing, whereas others remain crash free throughout their driving careers. Until fairly recently, little has been known about which behavioral and physiological changes differentiate competent older drivers from those experiencing a decline in driving ability. However, numerous studies in the literature (e.g., Ball, Owsley, Sloane, Roenker, & Bruni, 1993; Ball et al., 2002; Goode et al., 1998; Owsley et al., 1998; Owsley, Ball, Sloane, Roenker, & Bruni, 1991; Rizzo & Dingus, 1996; Rizzo, Reinach, McGehee, & Dawson, 1997) demonstrate that the driving performance of older drivers is related to a specific functional measure of visual processing speed, the useful field of view test (UFOV®). Furthermore, Ball and her colleagues (e.g., Ball, Ball, et al., 1988; Ball & Owsley, 1991; Edwards et al., 2002) have

shown that the size of the useful field of view can be expanded with training. Therefore, the present investigation was designed to determine if speed-of-processing training also produces an improvement in driving performance. Given that 17% (55 million) of the population is expected to be over age 65 by the year 2020 (U.S. Department of Transportation, 1997), the development of techniques to help older adults maintain their mobility and lessen the risk of crash involvement has clear societal benefits.

The *useful field of view* has been defined as the area from which one can extract visual information in a single glance without eye or head movement (Sanders, 1970). It is measured binocularly and can involve the detection, localization, or identification of targets against more complex visual backgrounds (Verriest et al., 1983, 1985; Sanders, 1970). The size of the useful field of view is affected by many factors,

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